COST EVALUATION EVENT

Foreword:

In real life, preparing an accurate bill of materials and cost accounting gives engineers clarity on the number of components used, time taken for production, and cost per part calculation. Such classification allows for control of cost and ascertaining the profitability of processes, which helps make strategic decisions to manufacture parts more efficiently.

The BAJA SAEINDIA's cost event is an effort to put students through the same thought process and help students learn the nuances of cost accounting and the benefits of maintaining a bill of materials in the product development process.

Objective:

Cost is one of the most critical aspects of any commercial entity, but the event here is to not only report the most optimum cost of the component in the prototype or production stage but instead focus on how well the team understands what has gone into their part and applying their learning of procurement and manufacturing techniques to optimize the cost, labor, time, material wastage and various overhead costs.

Cost Event consists of two related sections:

1. **Cost Report:** The cost report provides all the background information to verify the vehicle's actual cost and if the teams want to present any design features or fabrication processes that are innovative or are expected to result in significant cost savings.
2. **Prototype Cost:** The prototype cost is the actual cost of the fabrication of the vehicle and the points related there, too.

Scoring pattern:

EVENT	SCORE	PHASE
COST PRESENTATION	50	PHYSICAL ROUND (PHASE III)

Cost Report (Required Submission):

The Cost Report may contain a maximum of **Three** sections:

Report Section 1 - Overview (Required):

Cost overview submission is compulsory for all the teams. It is intended to allow each team to point out and briefly comment on any design features or fabrication processes that are innovative or are expected to result in significant cost savings. Teams may also use the overview to explain items or processes that might appear to be discrepancies within the report. The overview section is limited to a maximum of four (4) pages. This should be included as part of the Cost Documentation (pdf) file.

Report Section 2 - Cost Documentation:

This section includes copies of receipts, invoices, catalogue pages, online prices, or other documentation to substantiate the costs of the parts and materials of any item **costing more than Rs. 200.** Cost Documentation must be at full retail Indian prices.

The use of foreign receipts and purchases from discount sites such as Craig's List, eBay, company brochures/pricelists, or junk yards are not allowed. The report is expected to be comprehensive, well- documented, truthful, and accurate.

Report Section 3 – Engineering drawings:

This section includes the Engineering drawings (2D) of the components which are fabricated by the team. The drawing should consist of 3 basic views and important dimensions. The purpose of engineering drawing is to check the understanding of teams for optimum raw material sizing and manufacturing process selection. The drawing report should be composed sub-system-wise as given in the cost report template.

Prototype Cost (Required Submission):

Costing Sheets:

The core of the report is the series of costing sheets. This section must contain the one-page summary sheet broken up into the individual subsystems. A separate 'ALL OEM' sheet will contain all the procured components except bearings and fasteners (bearings should be included in their respective sub-system, and all the fasteners should be included in the 'Fasteners' sheet). Each subsystem needs an individual subassembly sheet, which will include the fabricated parts. Note that Vehicle Assembly Labor cost is for the labor it takes to assemble a subassembly to the frame. Note that the sub-system assembly time is the time it takes to assemble all the parts in that assembly.

The team can download the costing sheet template from the BAJA SAEINDIA Forum & Website. Using online platforms such as Google Sheets / SharePoint may lead to errors. Hence, teams should strictly use the MS Office Excel Desktop app in offline mode to edit the Cost report template.

OEM Component cost library:

- This year, in the cost event, Teams are required to use the cost of the OEM components mentioned in the cost library.
- If any OEM component used by the team is not listed in the library, then the team shall fill up the Google form released by BAJA SAEINDIA and provide the cost of that OEM Component with appropriate bills in the required period. The team can fill out the Google form multiple times if two or more OEM components are not listed in the library.
- If the component is already listed in the Cost library, then the team should not fill out the Google form even if the price available at the team's end for any component is less than the price mentioned in the library.
- The new cost report template with an updated OEM component cost library will be released by **5th Dec. 2025** at the **BAJA SAEINDIA Forum**.

Google form link: BSI25_OEM_COST_SUBMISSION Deadline: 1st Dec. 2025, 23:59 IST

List of OEM Components strictly to be used from the Cost library:

- **Suspension:** Shock absorbers, Tires & Rims.
- **Brakes:** Calipers, Master cylinder.
- **mBAJA Powertrain:** Engine, Fuel tank.
- **mBAJA Drivetrain:** CVT.
- **eBAJA Powertrain:** Battery HV, Motor, Controller, Throttle pedal.
- **hBAJA Powertrain:** Engine, Fuel System, CNG Tank, CNG Kit, Electrical Harnesses.

Note:

- For sponsored components, if the cost of the components is lower compared to others, BAJA SAEINDIA reserves the right to adjust the cost.
- Bills / Invoices of the above-mentioned OEM components are required.
- If any of the above components are designed and manufactured by the team or designed by the team and manufactured by 3rd party, then it will not be considered as an OEM component.
- For the OEM Component listed above, if any deviation is found between the cost mentioned in the cost library and the cost given by the team, then the team will be penalized.
- For eBAJA teams the cost for the additional battery pack utilized for swapping during the event is to be omitted from the Overall Vehicle Cost.
- This document should not be modified from its current form, this includes password protection and embedding macros. Strictly open the document with Microsoft Excel (Offline Version). Teams will receive zero (0) points for Cost if the report is in the incorrect format or the files have been modified.

Penalties:

- If the judges determine that a team's cost estimates are inadequate and manipulated to lower expenses, the cost of the incorrectly estimated components will be tripled and added to the total cost summary.
- Submission of a Dummy/ False Cost Report will lead to a 25-mark deduction from Cost Presentation (Physical round).

Disclaimer:

The cost of the above-mentioned OEM components in the library are indicative figures to have uniformity amongst all the teams and should not be considered as an actual quote from the vendor / Supplier. The Supplier / Vendor names listed in the Cost library should not be considered as recommended vendors from BAJA SAEINDIA.

Cost Component Categories:

Teams must put items that are specified in the correct component categories and subcategories, or the items will not be considered. See the Cost Template for more details on component categories.

Cost Report – Submission Format:

Submission Link	BAJA SAEINDIA Website To be done on the BAJA SAEINDIA website in the submission section in the team profile.		
Last Date of Submission	**27th Dec. 2025, 23:59 IST**		
Document	Submission	File Type	File Name
Costing Sheet	Compulsory	.xlsx/.xls	TeamID_Cost Report.xlsx
Cost Documentation, Cost Overview	Compulsory	.pdf	TeamID_Cost Documentation.pdf
Engineering Drawings	Compulsory	.pdf	TeamID_Engineering_Drawings.pdf

Note: The team should ensure that while submitting the report sheet, it is in either. xlxs/.xls (no macros) extension.

On-Site Evaluation:

Objective:
The Physical round of the cost evaluation event has been introduced to evaluate the accuracy of the team's documentation and the adjustments made to the prototype cost, if any, after the fabrication of their vehicle.

Eligibility:
All Teams with submissions will be eligible to participate in the Cost Event at the physical event site.

Time Duration:

The evaluation will be done over a period of 30-minute time (5min for Change over + 15min for Presentation + 10min for Q&A).

Members Requirement:

A maximum of 5 members are allowed during the cost evaluation.

Required Documents:

Teams are required to be present with their vehicles in the cost evaluation bay during their respective slots with the hard copies of the following document:

1. **Overview**
2. **Costing Sheet**
3. **Cost Documentation**
4. **Engineering Drawing**
5. **Cost Adjustment Form -** The cost adjustment form allows teams to

mention any deviations from the prototype cost submitted in the Virtual event. Teams need to bring the cost adjustment form duly filled if the team would like to bring to the notice of judges any deviation from prototype cost submitted earlier.

6. **A1 Size Posters -** Teams are required to carry A1 Size Poster supporting the points from overview for on-sight evaluation.

Team can download the cost adjustment sheet from the below mentioned link:

https://bit.ly/BSI26_Cost_Adjustment_Form

Evaluation Criteria:

- Teams will be evaluated for a total of 50 points.
- The points awarded to teams during the Cost Event will be added to the score obtained for the final ranking.
- Teams will be scored based on the following criteria:

1. Final Prototype Cost

Cost Adjustment by teams:

The purpose of the cost adjustment form is to make additions to the previously submitted report. Items may be deleted, but the total adjustment for the individual component categories must be positive (cost will not be subtracted). This gives the team the chance to add items that were not previously planned. It is not an opportunity to redo the entire report.

- The total amount of adjustments may not exceed 10% of the total cost of the vehicle previously submitted.
- If the adjustment exceeds 10%, the additional amount will be added with a multiplier of two times(2x) of the additional amount.
- If the adjustment exceeds 25%, the report will be considered as excessive adjustment and the team will be disqualified from the event.
- Teams need to bring a hard copy of the Cost Adjustment Form during the Cost Event.

Cost correction done by judges:

- The cost sheet is to be judged based on its accuracy to the components reported and the team's understanding of the report. The report will be reviewed for cost accuracy regarding the material/process tables alongside the engineering drawing provided and to ensure that the teams have followed all the procedures and report guidelines
- If the judges find that team has excluded the cost of some components or has provided inadequate/unjustified costing with the intention to manipulate the overall cost, the judges will add an adjustment amount to correct the costing of the prototype.
- **The cost of inaccurate estimated components will be adjusted by the**

judges and the adjustment will be tripled and added to the overall vehicle cost.

- Once, the evaluation is done, the initial prototype cost, cost adjustment done by team and cost correction done by judges will be added.

2. **Presentation and Cost Report Evaluation:**

Teams must bring A1-sized posters for on-site evaluations. Font

Style: Calibri & Font Size: 14 or above.

The posters should comprehensively address the following points but are not limited to:

- Comprehensive Reporting-Team must report all processes and parts utilized in the design.
- Teams to give detailed explanations for the components used.
- Cost-effectiveness of part production technique/methods.
- The ability of a team to address and participate in discussions with the judges based on the questions raised by the panel.
- Implementation of manufacturing techniques reducing the production lead time.
- Impact of the Cost on Design and Material Selection.
- Recyclability and Environmental Impacts.
- Ability to meet Engineering Specifications & Standards**.**

<u>Judging Criteria (points to be included but not limited to):</u>

The cost sheet is to be judged based on its accuracy to the components reported and the team's understanding of the report. The report will be reviewed for cost accuracy regarding the material/process tables and to ensure that the teams have followed all the procedures and report guidelines.

- Comprehensive reporting-Team must report all processes and parts utilized in the design.
- Teams to give detailed explanations for the components used.
- Cost-effectiveness of part production technique/methods.
- The ability of a team to address and participate in discussions with the judges based on the questions raised by the panel.
- Implementation of manufacturing techniques by reducing the production lead time.
- Impact of the Cost on Design and Material Selection.
- Recyclability and Environmental Impacts.
- Ability to meet Engineering Specifications & Standards.